



INDIAN INSTITUTE OF INFORMATION TECHNOLOGY, SRI CITY
UG1 - IC - Signals & Systems - Scheduled Quiz – SPRING 2026

Date: 13-04-2026

Duration: 30 Mins (5:40 PM - 6:10 PM)

Max. Marks: 15

Instructions: Each question carries 1 mark. The answer option must be written within the square brackets/blanks, overwriting will lead to wrong marking

Name:

Roll No.:

Sec:

Room No.:

If the Fourier transform of $x(t)$ is, $X(j\omega) = \frac{j\omega}{(1+j\omega)^2}$, determine the Fourier Transform of $x(2t)$..... a. $\frac{j\omega}{(1+j\omega)^2}$ b. $\frac{j2\omega}{(2+j\omega)^2}$ c. $\frac{j\omega}{(2+j\omega)^2}$ d. None of these	[]
What is the inverse Fourier transform of $2\pi\delta(\omega + \omega_0)$ a. $e^{j\omega_0 t}$ b. $e^{-j\omega_0 t}$ c. $e^{j\omega t}$ d. None of these	[]
If the Fourier coefficients of a continuous time Fourier Series are, $a_1 = a_{-1} = 1/2$ and $a_3 = -a_{-3} = j/4$ and the other coefficients are zeros. Determine the signal $x(t)$ with $\omega_0 = 1$ a. $\cos t - \frac{1}{2}\sin 3t$ b. $\cos t + \frac{1}{2}\sin 3t$ c. $\cos t - \frac{j}{2}\sin 3t$ d. $\cos t + \frac{j}{2}\sin 3t$	[]
For the given Fourier coefficient $a_1 = a_{-1} = 1/2$ and $a_3 = -a_{-3} = j/4$ of a Fourier Series, determine the average power present in the signal per one time period..... a. 3/2 b. 5/16 c. 5/8 d. None of these	[]
If a continuous time periodic signal $x(t)$ with time period T, then the sum of the average powers in all of its harmonic components is equal to a. $\int_0^T x(t) ^2 dt$ b. $\int_0^T x(t) dt$ c. $\frac{1}{T} \int_0^T x(t) ^2 dt$ d. None of these	[]
If a continuous time signal $x(t) = e^{-t/2}$ for $0 < t < \pi$ in the fundamental interval with time period $t = \pi$; Determine the real and imaginary components of the Fourier Coefficients a. $real(c_k) = 0.6732 \frac{1}{1+4k^2}$; $imag(c_k) = 0.6732 \frac{-j2k}{1+4k^2}$ b. $real(c_k) = 0.5043 \frac{1}{1+16k^2}$; $imag(c_k) = 0.5043 \frac{-j4k}{1+16k^2}$ c. $real(c_k) = 0.4389 \frac{0.5}{0.25+4k^2}$; $imag(c_k) = 0.4389 \frac{-j2k}{0.25+4k^2}$ d. $real(c_k) = 0.3143 \frac{0.5}{0.25+k^2}$; $imag(c_k) = 0.3143 \frac{-jk}{0.25+k^2}$	[]
Using the Dual of integration in time property determine the inverse Fourier transform of $H(\omega) = \int_{-\infty}^{\omega} \frac{2a}{a^2 + \eta^2} d\eta$ a. $jte^{-a t }$ b. $jte^{-a t } + \pi\delta(t)$ c. $-\frac{e^{-a t }}{jt} + \pi\delta(t)$ d. $\frac{e^{-a t }}{jt} + \pi\delta(-t)$	[]



INDIAN INSTITUTE OF INFORMATION TECHNOLOGY, SRI CITY
UG1 - IC - Signals & Systems - Scheduled Quiz – SPRING 2026

Determine the Fourier transform of the signal $h(t) = \frac{4t}{(1+t^2)^2}$ a. $2\pi e^{- \omega }$ b. $\frac{2\pi e^{- \omega }}{j\omega}$ c. $\frac{2\pi e^{- \omega }}{j\omega} + \pi\delta(\omega)$ d. $j2\pi\omega e^{- \omega }$	[]
Determine the Fourier coefficients of the signal $y(t) = x(t)e^{jM\omega_s t}$ where the periodic signal in the fundamental interval $x(t) = A$ in $-\frac{T_1}{2} < t < \frac{T_1}{2}$ and zero in the rest of the interval. Assume the time period of $x(t)$ as T. a. $\frac{\sin[(k-M)T_1/2T]}{\pi(k-M)/T}$ b. $\frac{\sin[(k-MT_1)/2T]}{\pi(k-MT_1)/T}$ c. $\frac{\sin[kT_1/2T-M]}{\pi k/T-M}$ d. $\frac{\sin[kT_1/2MT]}{\pi k/MT}$	[]
Given that the Fourier coefficients $c_k = \frac{a}{2} \frac{1-e^{-jkn}}{jkn}$ of a square wave with time period $2T$ and non-zero value ($= a$) of duration T in the fundamental interval. Determine the Fourier coefficients of the ramp signal again with time period $2T$ defined as $y(t) = at$ for $0 < t < T$ and zero otherwise. a. $aTe^{-jkn/2} \frac{\sin(kn/2)}{k^2\pi^2}$ b. $aTe^{-jkn} \frac{\sin(kn/2)}{k^2\pi^2}$ c. $aTe^{-jkn/2} \frac{\sin(kn/2)}{jk^2\pi^2}$ d. $\frac{aT}{2} \frac{1-e^{-jkn}}{jk^2\pi^2}$	[]
A seismic signal is defined as $k(t) = \text{rect}(\frac{t+1}{2})$. The Frequency domain representation of the signal is? (a) $2 \text{sinc}(\omega)e^{j\omega}$ (b) $\text{sinc}(\omega/\pi)e^{-j\omega}$ (c) $3 \text{sinc}(3\omega/\pi)e^{j\omega}$ (d) $2 \text{sinc}(\omega/\pi)e^{j\omega}$ (e) None of them	[]
If signal $g(t) = \begin{cases} 1, & t > 0 \\ 0, & t = 0 \\ -1, & t < 0 \end{cases}$, finds the Fourier transform of the signal $\frac{dg(t)}{dt}$? (a) 1 (b) 0 (c) 1.5 (d) 0.5 (e) 2.5 (f) None of them (ans. 2)	[]
The energy of the signal $y(t) = 4 \text{sinc}(t/5)$? (a) 30 (b) 60 (c) 70 (d) 80 (e) None of them	[]
If $x(t)$ is a real valued and periodic signal. The Fourier-series coefficient of $x(t)$ is c_p. The following which is true? (a) $-c_p = c_p^*$ (b) $c_p = - c_{-p}^*$ (c) $c_p^* = c_{-p}$ (d) $\text{Real}(c_p) = -\text{real}(c_p^*)$ (e) None of them	[]
Given $x(t) \xrightarrow{FS, T \rightarrow 0} c_k^x = \begin{cases} j4k, & k < 4 \\ 0, & \text{else} \end{cases}$. If $y(t) = \int_{-\infty}^t x(t-8)dt$, then Fourier-series coefficient c_1^y of $y(t)$ is.....? (a) $8/\pi$ (b) $32/\pi$ (c) $24/\pi$ (d) $14/\pi$ (e) None of them (ans. $16/\pi$)	[]